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The Liverpool Pedestrian project was focusing more on developing an edge computing visual sensor in order to monitor pedestrians walking down the streets, bicycles and even vehicles while also maintaining their privacy. The goal here is to transform existing CCTV infrastructure to intelligent traffic systems that support data driven planning. As their central business district expanded, city planners should use better tools to get more knowledge on mobility patterns without violating privacy laws without expensive infrastructures

This project uses an embedded device called Jetson TX2 which was made by NVIDIA. This was combined with YOLOv3 which is an object that detects algorithms with SORT tracking. By using a video directly on the device, metadata is being transmitted rather than raw video footage. The reason behind this was to reduce bandwidth usage and make sure privacy was being protected.

Oxford Town Center dataset shows an average of 69% on validation and processing speeds up to 20 frames per second. The system tends to underestimate detections in crowded areas, but it will still successfully capture overall traffic. A real world deployment was a system deployed inside a building during fire alarm. It can detect evacuation and confirm the building was cleared with success. Outdoor Deployment had twenty sensors deployed around Liverpool's CBD. This showed activity on pedestrian and traffics with highest peaks. Over 20,000+ objects were detected in a single day.

